Chatbot for Remote Physical Knee Rehabilitation with Exercise Monitoring based on Machine Learning

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***Abstract*— A machine learning-based chatbot has been developed for remote knee rehabilitation, incorporating exercise monitoring and feedback through video analysis. The MediaPipe framework is utilized for recognizing poses and movements of patients. Forty features have been constructed for machine learning algorithms to determine exercise characteristics. The model was trained using multi-class logistic regression and achieved an accuracy of 98% on test data. Video tests evaluated by the machine learning model were used for objective progress monitoring. The chatbot provides patients with feedback and objective progress indicators, facilitating controlled rehabilitation.**

Keywords-Remote rehabilitation, Chatbot, Knee, Feedback, Video analysis, Machine learning, Mobile application, Health and recovery.

I. Introduction

The problem of remote knee rehabilitation arises due to limited patient access to physical therapy and specialists for various reasons, such as remote location, time constraints, or financial issues. This can lead to inadequate or incorrect rehabilitation, delayed recovery, and knee re-injuries. Remote knee rehabilitation can be facilitated through telemedicine technologies, including the use of mobile applications, virtual trainers, video consultations, and other innovative solutions. However, while there are various approaches to remote rehabilitation using apps and messengers, they all have their limitations. These limitations include the lack of feedback on exercise performance, limited monitoring and correction capabilities, as well as restricted access to specialists.

In this study, we propose a solution to these problems by introducing video monitoring using a smartphone camera in the format of a Telegram chatbot. The chatbot acts as a virtual trainer, guiding the patient through an individual rehabilitation program and monitoring exercise execution. By employing machine learning algorithms, the system can analyze workout videos, accurately assess exercise technique, and identify potential errors and deviations.

II. Rate The Limitations of Chatbots in Physical Rehabilitation

There are various examples of using chatbots in physical rehabilitation. They are mainly focused on text-based communication with patients, providing recommendations, scheduling appointments, and other service-related functions [1]. Examples of using such chatbots for post-total knee arthroplasty rehabilitation are known [2]. It has been noted that this tool can be useful for improving adherence to home-based rehabilitation. Overall, chatbots for physical rehabilitation have some limitations related to the lack of feedback on exercise performance:

- Lack of precise assessment of execution: Most chatbots are based on providing instructions and video tutorials for exercises, but without feedback from the patient, it is not possible to determine if they are performing the exercise correctly.
- Limited control and monitoring: Without feedback on exercise performance, chatbots cannot effectively monitor the patient's progress, whether they are following the prescribed recommendations, or if they are experiencing any problems or pain.
- The effectiveness of a chatbot largely depends on the accuracy and completeness of the user-entered data and self-reports. However, patients may provide incomplete or inaccurate information, which can affect the quality of feedback and recommendations.

III. The Task Formulation

The concept of a feedback-enabled chatbot for rehabilitation combines machine learning, computer vision, and a chat interface to optimize the recovery process for patients.

The goal of this research is to contribute to the field of remote knee rehabilitation by creating an accessible tool that helps patients with knee injuries effectively and safely restore their functionality. The main focus of development is to provide patients with feedback based on video analysis and identify errors in exercise technique. Please note that the translation may not adhere to strict English grammar in order to maintain the meaning and context of the original text.

The functioning scheme of the feedback-enabled chatbot is illustrated in Figure 1.

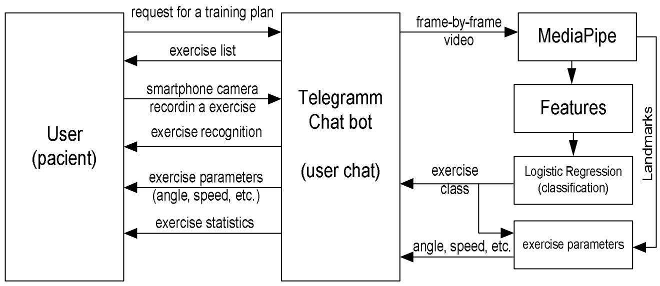

Fig. 1. Functioning scheme of the video-based feedback-enabled chatbot

According to the scheme, user interaction occurs through a text interface in a question-answer mode. The main functionality is revealed after uploading a video of exercise performance. The uploaded video is processed by the recognition module, where exercise features are extracted from each frame, independent of its size and scale. The obtained features are then used for classification procedures using a trained model. Based on the recognized exercise, angle measurements and motion speed parameters are calculated according to the target parameters. The final result is provided to the user as a text description and stored in the database for statistical tracking. Thus, to implement this chatbot scheme, the following tasks need to be addressed:

1. Choose a method for body pose keypoint detection and motion segmentation from video.

2. Define the exercise set and establish requirements for them.

3. Identify the main parameters.

4. Create a machine learning model for exercise classification and evaluation of performance quality.

5. Develop a method for objective assessment of recovery progress.

6. Design a chatbot interface that interacts with patients, accepts video recordings, provides feedback, and maintains statistics.

These tasks have determined the research design and a step-by-step approach to achieving the research goals. Let's consider the sequential implementation of each task.

IV. BODY POSE KEYPOINT DETECTION

To address the pose recognition task, the skeleton-based Human Pose Estimation (HPE) framework called MediaPipe was chosen. MediaPipe provides frame-by-frame detection of key landmarks on the patient's body, such as joints and limbs. The selection of MediaPipe over similar solutions was based on its good performance with video data, robustness in handling occlusions (referring to situations where parts or the entire body is hidden from the recognition system), and real-time optimization.

MediaPipe utilizes three pre-trained neural models to detect landmarks on the human body. For this work, the Full model variant was chosen as a compromise between recognition accuracy and speed, achieving an average of 21 frames per second (static mode = True, with default parameters).

To ensure reliable detection of body landmarks, the video should meet the following requirements:

- Clear visibility of the user's face throughout the recording.
- Absence of obstacles or occlusions that could hinder accurate landmark detection.
- Proper lighting conditions.
- A background that is not overly cluttered, and clothing that does not blend into the background.

Considering the specific requirements and the chosen parameters of the MediaPipe model for body landmark detection, the subsequent steps related to data collection for dataset formation need to be discussed.

V. EXERCISE SELECTION

To prepare the dataset, videos of 19 variations of rehabilitative exercises were recorded by 10 volunteers. Each video had a resolution of 1280x720 pixels and a frame rate of 30 frames per second. The framing of the videos was done in such a way that the volunteer, while lying down, occupied approximately 80% of the screen width, and while standing, occupied approximately 80% of the screen height. This chosen shooting mode provided sufficient information for MediaPipe and created optimal conditions for obtaining accurate data on the position and movement of the knee joints during exercise.

Each exercise in the set corresponded to specific requirements:

- The exercises had to be easy to perform and not require complex movements or acrobatic skills.
- The movements were selected so that the target joint movement occurred in a single plane.
- The movements had to be cyclical in nature, promoting muscle strengthening and increasing knee functionality.
- The exercises were chosen with a certain load on the knee joint to ensure a gradual increase in intensity and controlled loading during the recovery process.

Considering these requirements, a set of exercises was formed, consisting of five exercises performed in a seated position, five exercises performed in a standing position, and five exercises performed in a lying position. This variety of poses and positions allows for the engagement of different muscle groups and joints during rehabilitative exercises. Seated exercises provide a stable base for working with the knee joint. Standing exercises contribute to strengthening the lower limbs and balance. Lying exercises allow for working with the knee joint in a more relaxed state. Performing exercises for both limbs also ensures symmetrical muscle and joint development, which is important for functional recovery and balanced rehabilitation [3].

VI. FEATURE CONSTRUCTION

As part of this task, the coordinates of the landmarks were transformed into meaningful attributes to obtain the necessary information about the body pose. The output of MediaPipe consists of landmarks, and each landmark has four parameters: x position ϵ [0,1], y position ϵ [0,1], depth z ϵ [0,1], visibility ϵ [0,1], and presence ϵ [0,1]. In this work, we use the x, y, and z parameters. The depth parameter (z) was taken as is. The x and y parameters were transformed into scale-independent values.

Firstly, joint bending angles were calculated. The angle "pao" between three points P, A, and O can be calculated using the following formula:

$$pao = \arccos((AP^2 + OP^2 - AO^2)/(2 \cdot AP \cdot OP)), \quad (1)$$

where AP represents the distance between points A and P, and OP represents the distance between points O and P.

Distances between landmarks were also computed to reflect the spatial relationships between body parts. The distance "dab" between two points a and b can be calculated using the formula:

$$dab = \sqrt{((ax - bx)^2 + (ay - by)^2)}, \quad (2)$$

where (ax, ay) and (bx, by) represent the coordinates of points a and b, respectively.

As a result of applying these transformations, a dataset was created, consisting of 40 features comprising 14 angles (a), 16 distances (d), and 10 depths (z). In addition to the attribute features, the dataset included a target attribute representing the exercise class and a textual description attribute providing contextual information about the performed exercise.

The attributes and target information were obtained for each frame in the dataset, resulting in a dataset size of 42 by 693025 for 10 individuals. This dataset then served as a resource for subsequent training.

To solve the classification problem in the context of this dataset, the logistic regression model was used. It has been well-known for its effectiveness in binary and multiclass classification tasks. In this case, with 19 classes (exercises) in the dataset, logistic regression provides a suitable framework for multiclass classification.

The hyperparameters of the model were chosen based on a balance between model complexity, regularization, and prediction efficiency. During the training process, it was observed that the logistic regression model exhibited signs of overfitting on the training data. To address this issue, regularization parameters were introduced. In this implementation (using scikit-learn), L2 regularization (ridge regression) with a parameter C=0.1 and the "liblinear" optimization algorithm were selected. The model was trained with strong regularization, which helped mitigate overfitting to some extent.

The dataset was split into 70% for training data and 30% for testing data, ensuring that the model's performance is evaluated on unseen data to reliably assess its generalization capabilities. After training the logistic regression model on the training data, an accuracy of 98% was achieved. Figure 2 shows the corresponding confusion matrix.

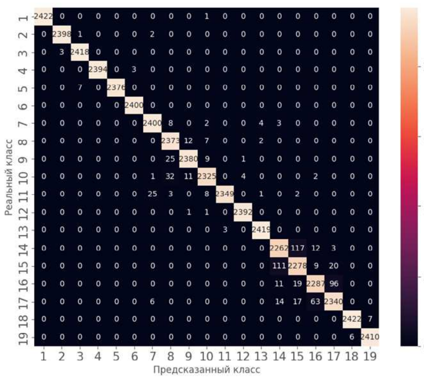

Fig. 2. Confusion matrix of the logistic regression model for 19 exercises

To assess the model's effectiveness on unseen data, the trained logistic regression model was evaluated on the test data. Remarkably, the model achieved an accuracy of 98% on the test set as well.

VII. Joint Activity Parameters

The following parameters were defined for each exercise class to analyze the characteristics of the target movements:

- Minimum flexion angle (min_angle).
- Maximum flexion angle (max_angle).
- Range of motion, calculated as the difference between the maximum and minimum angles (swing = max_angle - min_angle).
- Mean angle (mean).
- Standard deviation (std_dev) of the angle values, indicating the spread of angle values from the mean.
- Instantaneous angular velocity.
- Average angular velocity.
- Maximum angular velocity.
- Time to reach maximum angular velocity (t_max_omega).
- Mean acceleration.

These parameters serve as important tools to assess the effectiveness and quality of exercise performance within a specific exercise class and throughout the entire rehabilitation training. They are crucial for accumulating statistics and tracking the progress of rehabilitation. By comparing these parameters before and after training or rehabilitation sessions, an objective evaluation of the rehabilitation program's effectiveness can be made.

VIII. Chatbot Interface

To implement the chatbot interface, a prototype was developed. Upon initialization, the chatbot greets the patient, establishing a friendly and informative tone. It then presents a menu of available exercises, allowing the patient to select an exercise they would like to perform. Detailed instructions on how to properly perform the chosen exercise are provided to the patient. An example conversation with the chatbot is shown in Figure 3.

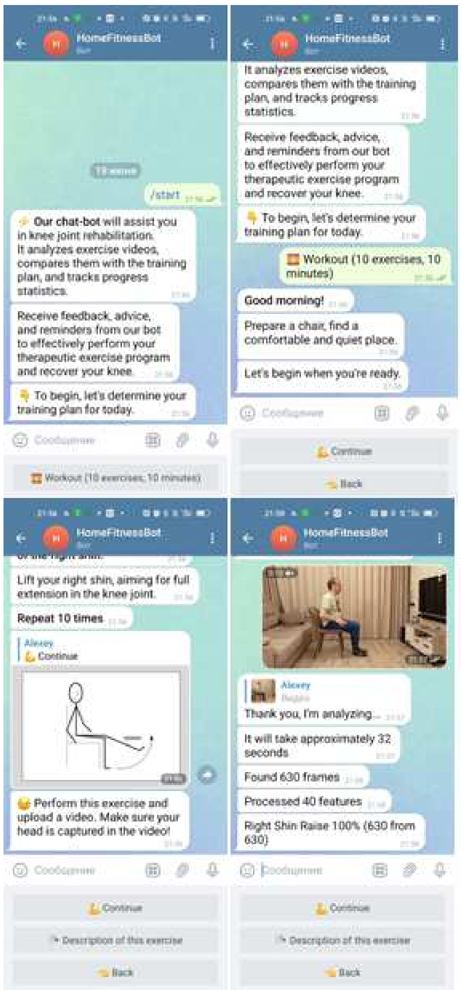

Fig. 3. Placeholder for the example conversation with the chatbot

To assess the execution of the exercise, the algorithm prompts the patient to record a video of the exercise using a compatible device. The recorded video is then preprocessed using the MediaPipe framework, which extracts relevant features, including body pose and joint angles.

To track progress, the algorithm maintains a comprehensive history of the patient's exercise sessions [3]. This includes recording the number of repetitions, duration, and overall progress over time. Each interaction with the interface concludes by offering the patient the option to continue with another exercise or end the session. Patient engagement and autonomy are encouraged, ensuring a user-centered approach to remote rehabilitation.

IX. Limitations

During the testing of the proposed solution, several limitations and drawbacks were identified, which restrict the usability of this approach. These include:

1. Lack of consideration for the physical environment: The proposed implementation of video workouts with a chatbot does not take into account the physical environment in which rehabilitation exercises are performed. Factors such as limited space, presence of obstacles, or the suitability of the surrounding environment for safe and effective rehabilitation may hinder the rehabilitation process.

2. Ethical and legal considerations: The requirement to upload workout videos that may reveal the patient's home environment and their face raises ethical considerations and legal norms related to patient confidentiality, data security, and responsibility.

3. Technical limitations of the Telegram Bot API: There is a technical limitation in the Telegram Bot API that restricts the size of uploaded files in the chat to 20 megabytes per bot. Future plans may involve transitioning to the MTProto protocol to overcome this limitation.

It is essential to consider these limitations when implementing similar chatbot solutions in medical practice. Addressing these issues may involve obtaining feedback from patients and integrating with healthcare professionals to ensure a more comprehensive and compliant approach.

X. Conclusion

The proposed study has the potential to overcome the limitations of traditional approaches to remote knee joint rehabilitation by leveraging the capabilities of chatbots and machine learning. By providing personalized exercise monitoring and adaptive feedback, the developed system can optimize the rehabilitation process, enhance patient engagement and motivation, and potentially reduce the need for in-person therapy sessions. This research contributes to the advancement of remote rehabilitation technologies and aims to improve the accessibility and effectiveness of knee joint rehabilitation programs.

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